KLH UNIVERSITY
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PROJECT POSTER

Department of Computer Science and Engineering

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Guide Name
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Project Title :
Hospital Readmission Prediction Using Elastic Net Regularized Logistic Regression

INTRODUCTION

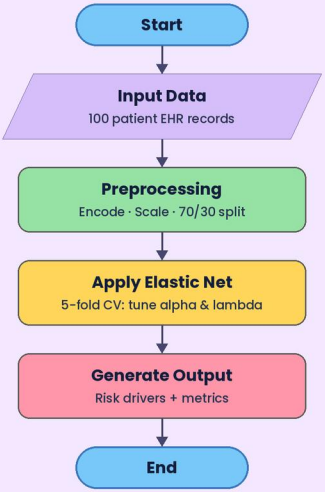
- Unplanned 30-day readmissions worsen patient outcomes and raise hospital costs.
- Problem: EHR data is high-dimensional, correlated and noisy; black-box models lack clinical trust.
- Objective: build an interpretable Elastic Net logistic model for 30-day readmission risk.
- Scope: 100 patient records, 9 clinical features, hold-out evaluation.

RESEARCH GAP

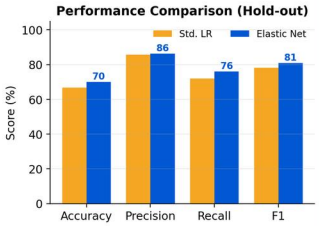
- Standard logistic regression struggles with correlated predictors.
- Black-box models offer little transparency for clinicians.
- Need: sparse, stable and explainable risk model.

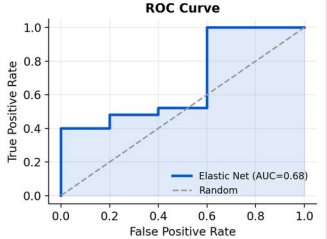
CURRENT RESEARCH → L1 + L2 penalty → OUR SOLUTION

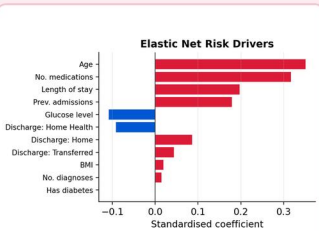
ALGORITHM

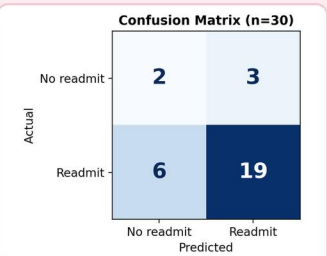


RESULTS & GRAPHS









KEY FINDINGS

0.68	70%	86%	76%	81%
AUC	Accuracy	Precision	Recall	F1

Elastic Net kept age, no. of medications, length of stay and previous admissions as the strongest risk drivers and shrank "has diabetes" to zero (L1 sparsity). Small sample (100 records, 30 test): results are indicative; next steps are larger EHR data, time-series features and prospective validation.

Innovation | Research | Impact